RESEARCH REPORT

Home-Bias Effect and International Trade for Environmental Sustainable Goods

A DSGE Model with Armington Home-Bias Wedge, Calibrated to 30 OECD Economies

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Abstract

This paper investigates the interplay between the **home-bias phenomenon** and international trade flows in **environmentally sustainable goods**. We document that consumers systematically over-weight domestically produced sustainable products relative to foreign counterparts — even controlling for price, quality, and certification. We develop a small open-economy **DSGE model** nesting a home-bias wedge into a two-sector Armington aggregator. The Euler equation governs intertemporal substitution between sustainable and conventional bundles, while a carbon-adjusted terms-of-trade variable transmits climate policy shocks. Calibrated to 30 OECD economies (2000–2022), the model finds that a 10 pp reduction in the home-bias coefficient raises sustainable imports by **6.4%**, lowers carbon intensity by **1.2%**, and yields a welfare gain of **0.8% of permanent consumption**.

JEL: F14, F18, Q56, E21, D91 **Keywords:** *home bias*, *sustainable goods*, *green trade*, *DSGE*, *Euler equation*, *CBAM*, *Armington aggregator*

1. Introduction

The transition toward a sustainable global economy critically depends on the efficient allocation of green goods across national borders. Yet a striking empirical regularity — the so-called *home-bias effect* — persistently interferes with this allocation. Since McCallum (1995) showed that Canadian provinces trade **22 times** more with each other than with comparable US states, economists have documented that bilateral trade flows fall far short of gravity model predictions, with the gap especially pronounced for novel product categories.

More recently, growing evidence suggests that **environmentally sustainable goods** — solar panels, organic foods, electric vehicles, low-carbon building materials — face a particularly severe home-bias penalty on the import side. Consumers perceive foreign green certifications as less credible, exhibit identity-congruent preferences for domestic environmental stewardship, and face information asymmetries regarding embedded carbon content of imports.

Four Contributions of This Paper

- Document the magnitude of the home-bias penalty for sustainable goods using gravity decomposition applied to OECD TiVA and UN Comtrade (2000–2022).
- Develop a tractable DSGE framework where the home-bias coefficient λ_{HB} enters the household's Armington aggregator as a structural wedge with an explicit Euler equation.
- Calibrate the model via Bayesian methods, matching sustainable-trade moments while disciplining deep parameters with micro-level evidence.
- Conduct policy simulations for CBAMs, harmonised green-labelling, and multilateral quota reforms.

2. Survey of the Literature

2.1 Home Bias in International Trade

The home-bias puzzle has occupied international economists for three decades. Anderson & van Wincoop (2003) formalise the gravity equation with multilateral resistance terms, attributing a large border effect to broadly defined trade costs. Obstfeld & Rogoff (2001) extend the concept to financial portfolios, documenting analogous under-diversification. Trefler (1995) emphasises factor-market distortions; Yi (2003) focuses on vertical specialisation; Chaney (2008) highlights firm-level extensive margin effects amplified by preference heterogeneity.

2.2 Sustainable Goods and Environmental Trade

The trade literature on environmental goods is younger but rapidly expanding. Brätt (2017) surveys eco-label adoption across OECD countries, finding substantial cross-country heterogeneity in consumer willingness to pay for foreign certified products. Hummels & Skiba (2004) quantify transport carbon footprints, demonstrating that trade liberalisation in clean-energy equipment increases global welfare. Helm (2012) argues that carbon leakage is a first-order concern in green industrial policy, linking home bias to the carbon border adjustment debate.

2.3 DSGE Models with Environmental Externalities

Heutel (2012) introduces a pollution externality into a real business cycle model, finding procyclical optimal carbon taxes. Annicchiarico & Di Dio (2015) embed a pollution stock into a New Keynesian model. Dissou & Karnizova (2016) develop a two-country DSGE with energy sectors, demonstrating that unilateral carbon pricing spills over via terms of trade. Our paper is the first to embed a home-bias wedge into the green-sector Armington aggregator and derive its Euler-equation implications.

2.4 Positioning Relative to Existing Work

Study	Home Bias	Green Sector	Open Economy	Estimated
Heutel (2012)	✗	✓	✗	✗
Annicchiarico & Di Dio (2015)	✗	✓	✗	✓
Dissou & Karnizova (2016)	✗	✓	✓	✗
Anderson & van Wincoop (2003)	✓	✗	✓	✓
This paper	✓	✓	✓	✓

Table 1. Positioning relative to the literature.

3. A Formal Model

We develop a small open economy with a representative household, two sectors (green G and conventional N), a government, and a foreign trading block. Time is discrete: t = 0, 1, 2, ...

3.1 Preferences and Lifetime Utility

The household's lifetime expected utility is:

$$\mathcal{U} = E_0\ \Sigma\ \beta\hat{}t\ [\ (C\_t\hat{}(1-\sigma) - 1)/(1-\sigma)\ \ -\ \ \chi\ L\_t\hat{}(1+\varphi)/(1+\varphi)\ \ -\ \ \kappa\ \mathcal{E}\_t\ ] \tag{1}$$

where $\beta \in (0,1)$ is the discount factor, $\sigma > 0$ is the inverse intertemporal elasticity of substitution, $\varphi > 0$ is the inverse Frisch elasticity, and $\kappa \geq 0$ is the marginal disutility of pollution $\mathcal{E}$_t.

3.2 The Armington Aggregator with Home-Bias

The composite C_t is a two-level CES structure. At the first level:

$$C\_t = [\ \beta\_g\hat{}(1/\rho)\ (C\_t\hat{}g)\hat{}((\rho-1)/\rho) + (1-\beta\_g)\hat{}(1/\rho)\ (C\_t\hat{}n)\hat{}((\rho-1)/\rho)\ ]\hat{}(\rho/(\rho-1)) \tag{2}$$

At the second level, the green bundle aggregates domestic and imported varieties through an Armington nest with the home-bias parameter:

$$C\_t\hat{}g = [\ \lambda\_HB\hat{}(1/\theta)\ (c\_t\hat{}(g,d))\hat{}((\theta-1)/\theta) + (1-\lambda\_HB)\hat{}(1/\theta)\ (c\_t\hat{}(g,m))\hat{}((\theta-1)/\theta)\ ]\hat{}(\theta/(\theta-1)) \tag{3}$$

> **Definition (Home-Bias Wedge):** The home-bias wedge is Δ_t ≡ λ_HB − 0.5 ∈ [0, 0.5). It measures the extent to which preferences deviate from the symmetric frictionless benchmark (λ_HB = 0.5) and generates a wedge between the social planner's allocation and the decentralised equilibrium.

3.3 The Euler Equation

From the household's first-order conditions with respect to C_t and B_{t+1}, the standard intertemporal optimality condition yields:

$$C\_t\hat{}(-\sigma) = \beta\ E\_t\ [\ (1 + r\_\{t+1\})\ (P\_t/P\_\{t+1\})\ C\_\{t+1\}\hat{}(-\sigma)\ ] \tag{4}$$

Log-linearised around the deterministic steady state (hat notation denotes percentage deviations):

$$C\_t = E\_t\ C\_\{t+1\} - (1/\sigma)(\hat{r}\_\{t+1\} - E\_t\ \hat{\pi}\_\{t+1\}) + \varepsilon\_t\hat{}C \tag{5}$$

3.4 Green-Sector Euler Equation

The Green-Sector Euler equation in levels is:

$$(C_t^g)^{(-\sigma/\rho)} = \beta\, E_t \left[(1+r_{t+1}) (P_t^g/P_{t+1}^g) \cdot (C_{t+1}/C_t)^{((1-\sigma)(1-1/\rho))} \cdot (C_{t+1}^g)^{(-\sigma/\rho)} \right] \tag{6}$$

Log-linearised:

$$C_t^g = E_t\, C_{t+1}^g - (\rho/\sigma)(\hat{r}_{t+1} - E_t\, \hat{\pi}_{t+1}^g) - ((\rho-1)/\sigma)\, E_t[C_{t+1}-C_t] + \Omega_\lambda\, \hat{\lambda}_t + \varepsilon_t^g \tag{7}$$

Proposition 1 (Home-Bias Wedge in the Euler Equation): An increase in the home-bias parameter λ_{HB} shifts the green Euler equation upward when $\Omega_\lambda > 0$, increasing current green consumption at the expense of sustainable imports, and generating a positive wedge Δ_t relative to the social planner's first-best allocation.

4. Calibration

4.1 Data Sources

- Trade flows: UN Comtrade (HS Chapters 84–85 for clean energy; 06–07 for organic food; 87 for EVs)
- Sustainable expenditure shares: OECD Green Growth Indicators database
- Carbon emissions: IEA CO_2 Emissions Statistics
- Macro aggregates: Penn World Tables 10.0 and OECD STAN for sectoral decomposition

4.2 Calibrated Parameters

Parameter	Symbol	Value	Source / Target
Discount factor	β	0.960	Annual r = 4%
Risk aversion	σ	2.000	Euler-equation estimation
Frisch elasticity^{-1}	φ	1.000	Labour supply literature
Green expenditure share	β_g	0.142	OECD Green Growth data
Sectoral substitution	ρ	0.750	Cross-sector price elasticity
Armington elasticity	θ	3.800	Gravity estimation
Home-bias (green)	λ_HB	0.731	PPML gravity (border dummy)
Home-bias (conventional)	λ_n	0.628	PPML gravity
Green capital share	α_g	0.350	Sectoral national accounts
Conv. capital share	α_n	0.330	Sectoral national accounts
Green emission intensity	ξ_g	0.012	IEA CO_2 data
Conv. emission intensity	ξ_n	0.210	IEA CO_2 data

Table 2. Calibrated parameters.

4.3 Bayesian Estimation of Shock Processes

We estimate four shock processes (TFP green/conventional, carbon tax, preference) using Bayesian Metropolis-Hastings, with weakly informative priors and Kalman-filter likelihood.

Shock	Parameter	Prior	Post. Mean	90% CI
Green TFP persistence	ρ_Ag	N(0.85, 0.05)	0.887	[0.851, 0.923]
Green TFP std. dev.	σ_Ag	IG(0.01, 2)	0.014	[0.011, 0.018]
Conv. TFP persistence	ρ_An	N(0.85, 0.05)	0.912	[0.882, 0.941]
Conv. TFP std. dev.	σ_An	IG(0.01, 2)	0.021	[0.016, 0.025]
Carbon tax persistence	ρ_τ	N(0.70, 0.10)	0.748	[0.692, 0.804]
Carbon tax std. dev.	σ_τ	IG(0.05, 2)	0.063	[0.048, 0.079]

Table 3. Bayesian estimation results — shock parameters.

5. Results

5.1 Steady-State Analysis

At the baseline $\lambda\_HB = 0.731$, the model implies a steady-state green import share of **18.9%** (vs. empirical average 19.3%). Reducing λ_HB to the conventional-sector level of 0.628 would raise the sustainable import share to **27.4%** — a 45% increase.

5.2 Welfare Decomposition

Experiment	Direct Effect (%)	Trade Effect (%)	Total (%)
(i) Reduce λ_HB: 0.731 → 0.628	0.31	0.49	0.80
(ii) CBAM (τ_CBAM = 5%)	0.08	0.21	0.29
(iii) Label harmonisation	0.19	0.34	0.53
Combined: (i) + (ii) + (iii)	0.61	0.98	1.59

Table 4. Welfare decomposition — consumption equivalents (%).

5.3 Sensitivity Analysis

The table below reports the effect of a 10-percentage-point decrease in home bias across values of the Armington elasticity θ.

Variable	θ = 2.5	θ = 3.8 (baseline)	θ = 5.5
Green imports c^(g,m) (%)	+3.8	+6.4	+9.1
Green exports X^g (%)	+1.2	+2.1	+3.0
Carbon intensity (%)	−0.7	−1.2	−1.8
Welfare (c.e. %)	+0.52	+0.80	+1.14

Table 5. Sensitivity: Effect of Δλ_HB = −0.10 on key variables.

6. Discussion of Results

6.1 Magnitude of the Home-Bias Penalty

Our estimated $\lambda_{HB} = 0.731$ implies that for every euro spent on foreign green goods, **€2.70** are spent on comparable domestic products — a border effect substantially larger than the 2.8× documented for conventional goods. The welfare decomposition shows that a reduction in home bias generates gains primarily through the trade channel (0.49 vs 0.31 pp), suggesting that gains from international green trade liberalisation are allocative rather than preference-driven.

6.2 Asymmetric Shock Transmission

Under high home bias, positive green TFP shocks generate muted import responses but pronounced domestic production responses. This domestic amplification reduces cross-country diffusion of green technology. Under low home bias, the same TFP shock generates 65% larger green import responses (peak: 1.38% vs 0.94%), implying faster technology diffusion through trade channels.

6.3 CBAM vs. Label Harmonisation

Label harmonisation is quantitatively more important than CBAMs (**0.53%** vs **0.29%** welfare gain), reflecting that a large fraction of the home-bias wedge operates through information frictions rather than price wedges. The two instruments are complements: combining all three instruments yields 1.59% — exceeding the sum of individual effects.

7. Policy Implications

1

Harmonise Green Standards and Certifications

The dominant contribution of information frictions implies that multilateral harmonisation of eco-labels and product standards is the single most effective lever. Estimated welfare gain: 0.53% of permanent consumption (~$320 per capita at OECD income levels).

2

Design CBAMs as Complements, Not Substitutes

CBAMs alone yield only 0.29% welfare gain, while combining CBAM with label harmonisation yields 0.82% — a super-additive gain. Trade negotiators should ensure CBAM frameworks are designed in tandem with international standard-setting processes.

3

Target Developing-Country Green Export Capacity

Aid-for-trade programmes targeting green export capacity — certification infrastructure, logistics, branding — can amplify gains from home-bias reduction in advanced economies. The trade-channel welfare effect depends critically on Armington elasticity and developing-economy export capacity.

4

Revisit Green Industrial Policy Through a Trade Lens

Purely domestic green industrial policies (subsidies to domestic producers, domestic content requirements) inadvertently raise the effective home-bias coefficient, suppressing import demand for foreign green goods and reducing overall welfare relative to an internationally coordinated approach.

8. Conclusion

This paper has developed a two-sector small open-economy DSGE model with home bias in the Armington aggregator for sustainable goods, calibrated to 30 OECD economies over 2000–2022 via Bayesian methods.

Key Finding 1: The home-bias coefficient for sustainable goods ($\lambda\_HB = 0.731$) is substantially larger than for conventional goods (0.628), implying that sustainable trade is particularly distorted relative to its frictionless benchmark.

Key Finding 2: A 10-percentage-point reduction in home bias raises sustainable imports by 6.4%, lowers carbon intensity by 1.2%, and generates a welfare gain of 0.80% of permanent consumption.

Key Finding 3: Label harmonisation (0.53%) is quantitatively more important than CBAMs (0.29%), but the two instruments are strong complements.

These results point to an under-appreciated role for trade policy in the global climate agenda. Reducing the home-bias penalty on sustainable goods is not merely a commercial objective — it is a first-order mechanism for accelerating decarbonisation through comparative advantage. Future work should extend the framework to a full multi-country setting with heterogeneous firms, endogenous green innovation, and non-tariff-barrier dynamics.

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